PROJECT PLAN

# DPP Data Pipeline

GROUP **RNN**

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To be served to:
PHILIPPINE MEDIA MONITORING LABORATORY (PMML)

PROJECT BLOG:
https://cs191-2627a-rnn.github.io

CS 191 WFR1
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# Customer Statement of Requirements

## I. Project Narrative

The Philippine Media Monitoring (PMM) Laboratory, based at the UP Diliman College of Media and Communication (CMC), studies how Filipinos talk about politics and elections on social media in their project entitled Digital Public Pulse (DPP). The Digital Public Pulse is a media monitoring effort that most recently tracked public conversations (i.e. social media posts) across Facebook, X (formerly Twitter), and YouTube for the entire 2022 election year and published its findings in a report during the same year.

The lab positions this research as useful to several groups. Government and election bodies can use it to shape policy that protects fair elections. Media organizations can see how their own coverage is spreading (or not spreading) once it reaches the public. Advocacy groups and civil society organizations can identify where to plug into the conversation. The findings also raise questions about whether platform companies (e.g. Meta, Google) are moderating election-related content in a way that keeps elections free and fair (Bunquin et al., 2022, pp. 12–13). All of this depends on first knowing what the conversation actually looks like, at scale, across platforms.

The 2022 findings illustrate this point. Over the course of the election year, mainstream news outlets lost ground in the conversation, while independent influencers, content creators, and ordinary accounts took on a larger role in shaping political discourse (Bunquin et al., 2022, pp. iii–iv). The lab treats each platform as a network of accounts connected through shares, replies, and recommendations. Groups of connected accounts, whether ordinary citizens, organizations, or influencers, form communities that behave like offline groups organized around a shared interest (Bunquin et al., 2022, p. 15). Tracking how those communities grow, shrink, or shift shows where political attention is going.

## II. Client's Previous Process and Methods

For the 2022 study, managing DPP meant building a list of more than 40 search terms across three categories: official election hashtags, the names of candidates and their campaign tandems, and the names of political parties (Bunquin et al., 2022, pp. 26–27).

In building that report, the PMM team pulled data from three platforms using three different tools. Facebook data came through CrowdTangle, accessed through a partnership with the news outlet Rappler, and was handed over in batches every few months. YouTube data came from a separate academic tool called YouTube Data Tools, a set of tools to extract YouTube video lists and channel network datafiles that connects directly with YouTube API v3. Twitter data came from a Microsoft Excel add-in called NodeXL Pro, filled in with a Python scraper for days the add-in could not reach.

Once collected, the data went through a separate cleaning step: posts linking outside the platform were removed, text fields were merged, punctuation and stop words were stripped, and non-English text was translated using the Google Translate API (Bunquin et al., 2022, pp. 29–30). Network measures were then calculated separately in a tool called Gephi, on a quarterly basis (Bunquin et al., 2022, p. 31).

That amounts to a keyword list, three separate collection tools, a separate cleaning step, and a separate analysis tool, repeated on a quarterly cycle. The report covers the full election year, May 2021 to May 2022, and its findings were published only after that entire year of data was collected and analyzed (Bunquin et al., 2022, p. i). A research process built around one big annual report does not show the lab what changed this month.

| METHOD | SPECIFIC TOOLS / PLATFORM | FREQUENCY / TIMING |
|---|---|---|
| Search Term Finding | Unspecified | Once; for the 2022 study |
| Facebook Data Collection | CrowdTangle | Quarterly in 2021-22 |
| YouTube Data Collection | YouTube Data Tools | Quarterly in 2021-22 |
| X/Twitter Data Collection | NodeXL Pro (MS Excel) + Python *snscrape* scraper | Regularly in 2021-22 |
| Data Cleaning | Google Translate API | After Data Collection |
| Data Storage / Organization | Google Drive | Unspecified |
| Data and Network Analysis | Gephi | After Data Cleaning |

Table 1: Client's Previous Data System

## III. Proposed Project Ideation

This project is planned to sit alongside the lab's research process. We plan to build a standardized pipeline that takes the same kind of platform data the lab works with, cleans and aggregates it using the same rules their methodology already applies, and shows the result on a dashboard the team can check on a regular schedule.

Pulling and processing this kind of data incurs real cost, both in terms of time and in what the underlying data services charge or restrict (monetary cost). For that reason, the system does not run in real time. It refreshes on a set schedule, likely daily or weekly, rather than continuously, so that the lab has a relatively recent picture without having to pay for constant data collection.

## IV. Requirements:

The proposed project will have the following list of requirements to consider based on previous context regarding the client's needs, as well as a preliminary meeting beforehand:

| NAME | DESCRIPTION |
|---|---|
| Manual Pipeline Trigger | The user can trigger a manual pipeline run on demand, in addition to any scheduled runs. |
| Dashboard View | The user can view the dashboard and see when the data was last refreshed. |
| Data Range Filter | The user can filter the data by a date range. |
| Data Export | The user can export the cleaned, aggregated data for use in deeper analysis. |
| Past Logs View | The user can see a log of past pipeline runs. |
| Data Cleaning | The system automatically cleans incoming data. |
| Data Aggregation | The system automatically aggregates cleaned data into summary counts. |
| Data Storage | The system automatically stores both the cleaned and the aggregated data. |

Table 2: Requirements

## Preliminary UI Design

The project is planned to have 3 points of access: logs, cleaned (processed) data, and then aggregated data. Currently, the logs are planned to be accessible through a dashboard with the help of spreadsheets in Google Sheets. The cleaned data is the raw data from the APIs and third-party platforms that has been treated and turned into usable data for aggregation. This portion is ideally accessible through a database or potentially in local data storage. Once aggregated, this data is what will be shown through the user dashboard, along with the logs.

Below is an AI-generated figure of this idea:

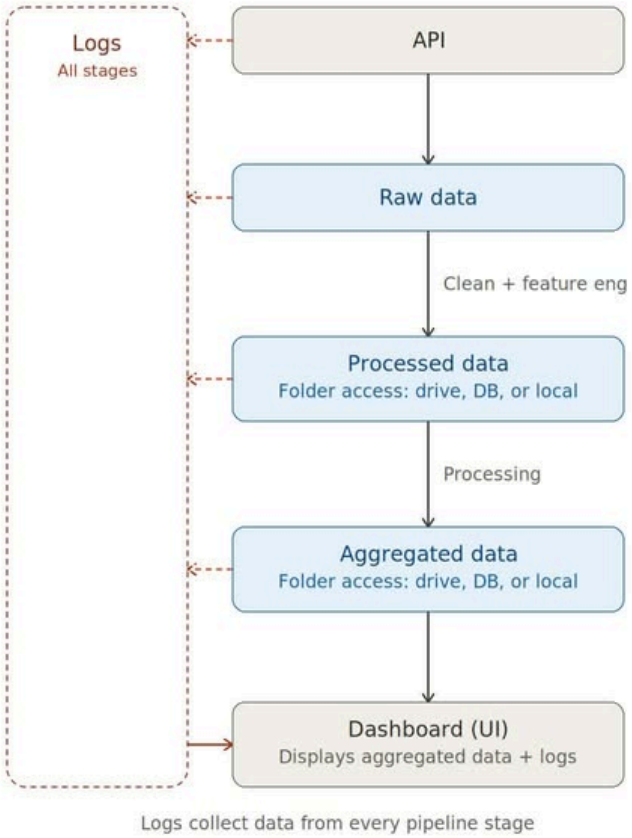



Figure 1: Preliminary UI Diagram (AI-generated)

# Plan of Work

## I. Estimate Cost and Resources

**Expected Financial Costs**:

| TOOL / PRODUCT | COSTS |
|---|---|
| Labor | P 0.00 *(Class Project)* |
| Third-party platform APIs | P 0.00 *(These will be handled by the client.)* |
| Database and Servers | P 0.00 *(Free Products will be used.)* |

Table 3: Expected Financial Costs (in Philippine peso)

**Labor Cost Estimate:**

| Task | Optimistic | Most Likely | Pessimistic | PERT Estimate |
|---|---|---|---|---|
| Client Communication | 30 h | 36 h | 42 h | 36 h |
| Project Proposal and Revisions | 10 h | 15 h | 22 h | 15 h |
| Prototyping | 16 h | 20 h | 28 h | 21 h |
| Project Development | 55 h | 70 h | 95 h | 72 h |
| Testing | 30 h | 40 h | 54 h | 41 h |
| **Total** | | | | **185 h** |

Table 4: Labor Cost Estimate (in man-hours)

**For System Resources:**

Developer Workstation / Computer: Any machine capable of running a modern code editor and Python

- Display: Color display for comfortable dashboard viewing, with a resolution of at least 1280x720
- Storage: Cloud storage of at least 100 GB
- Network: Stable broadband connection, at least 10Mbps download and upload speed

We note that these values are best-effort estimates, and will more than likely change, especially storage, as the specific nature and size of the data to be processed is determined.

## II. Identify Schedules and Milestones

Milestones the project will have to consider in the CS 191 portion include:

- November 11, 2026: Initial Software Release
- November 25, 2026: Software Test Report

| TASKS | W1 | W2 | W3 | W4 | W5 | W6 | W7 | W8 | W9 | W10 | W11 | W12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Meetings with Client | | | | | | | | | | | | |
| Initial Project Proposal | | | | | | | | | | | | |
| Prototyping | | | | | | | | | | | | |
| Project Proposal | | | | | | | | | | | | |
| Blog Updates | | | | | | | | | | | | |
| Project Development | | | | | | | | | | | | |
| Testing | | | | | | | | | | | | |
| **Milestones** | | | | | | | | | | | | |

Table 5: Project Schedule for CS 191 portion [September 7 - November 27, 2026]

## III. Project Metrics

To ensure the project status is transparent to the client, the following metrics will be utilized:

- **Schedule Variance:** planned vs actual completion date
- **Effort Variance:** estimated vs actual person-days per task
- **Pipeline Run Success Rate:** percentage of scheduled runs that complete without failure
- **Data Completeness Rate:** proportion of expected records successfully retrieved per run

## IV. Effort Breakdown

| RESPONSIBILITY | DESCRIPTION |
|---|---|
| Requirements & Design | Define data sources, target schema, pipeline architecture, and tooling choices based on the needs of the client. |
| Data Ingestion | Build connectors/extractors to pull posts/social media data via available APIs, including historical backfill, subject to rate limits, lookback restrictions, and budget constraints. |
| Data Cleaning and Transformation | Build an automated tool to handle missing values, type conversions, deduplication, as well as text preprocessing and enrichment such as sentiment and keyword tagging. |
| Pipeline Orchestration | Implement scheduling for periodic polling, dependency management, and workflow automation. |
| Testing and Validation | Unit tests, data quality checks, and end-to-end pipeline validation. |
| Deployment and Documentation | Deploy to production environment.<br>Write setup/maintenance documentations.<br>Provide ongoing support and bug fixes. |
| Communication and Client Coordination | Internal check-ins, status updates, and client meetings throughout the project. |

Table 6: Effort Breakdown

Our team consists of three members working under an open team structure. As such, rather than assigning fixed roles or dividing the project into separate ownership areas, we will approach most tasks collaboratively as a group. Work will be distributed organically on an ad-hoc basis based on availability, relevant skills, and the current needs of the project rather than predetermined assignments. Pair programming may also be employed.

To maintain accountability, the team will hold brief regular check-ins to review progress, surface blockers, and decide on next priorities. Tasks will be tracked on a shared board so that ownership and status remain visible to everyone, even as individual contributions shift.

## Project Risks

The RMMM (Risk Mitigation, Monitoring, and Management) included are some ways the team has included to potentially mitigate these risks before they occur.

| RISK | CATEGORY | PROBABILITY | IMPACT | RMMM |
|---|---|---|---|---|
| Third-party platform / API Disruption | DE | High | 3 | Maintain backups of collected data, and monitor third-party availability and API limits. |
| Insufficient Data Availability | DE | High | 3 | Use multiple data sources to reduce overreliance on sources. |
| Data Processing Errors | TE | Moderate | 4 | Create regular backups of processed data and test the data cleaner and aggregator for error prevention. |
| Data Pipeline Failure | TE | Moderate | 4 | Maintain regular data backups, and use version control to allow the pipeline to be reverted to a previous stable state. |
| Project Scalability | PS | Low | 2 | Design the project with scalability in mind by anticipating higher than usual influx of data and building sufficient capacity. |
| Sensitive Data Exposure and Misuse | BU | Low | 2 | Implement strict access-control protocols, ensuring only those with authorization can access sensitive data. |
| Unclear Requirements | CU | Low | 2 | Commit to regular communication with the client to clarify requirements, and document these agreements. |
| Priority Changes | PS | Low | 1 | Discuss changes in priority with the client and assess its feasibility before implementation. Utilize version control to revert to previous states if necessary. |
| Member Unavailability | ST | Moderate | 3 | Plan ahead for potential unavailability by sharing everything among team members so the project can continue. |

| | | | | |
|---|---|---|---|---|
| Communication with Client | CU | Low | 4 | Maintain regular communication with the client throughout the project, ensuring a positive working relationship. |
| Poor Estimates | PR | Moderate | 2 | Develop estimates using reliable data and methods, and review them before committing to these estimates. |
| Inadequate Documentation | PR | Low | 1 | Continuously document what occurs throughout the project process to be up-to-date. |
| Insufficient Resources | PS | Moderate | 3 | Use conservative estimates when planning for resource requirements, and include contingencies to account for increases in resource usage. |
| Poor Usability | CU | Moderate | 2 | Provide clear documentation and user guidance for the product, while maintaining regular communication with the client throughout development. |

Table 7: Project Risks

Legend (Based on Pressman, 2010):

Risk Categories:

- PS - Project Size
- BU - Business
- CU - Customer Characteristics
- PR - Process Definition
- DE - Development Environment
- TE - Technology to be Built
- ST - Staff Size and Experience

Risk Impacts:

- 4 - Catastrophic
- 3 - Critical
- 2- Marginal
- 1 - Negligible

# REFERENCES

Bunquin, J.B., Gaw, F., Baldo-Cubelo, J.T., Paragas, F., & San Pascual, M.R. (2022). *Digital Public Pulse: 2022 Philippine General Elections.* Philippine Media Monitoring Laboratory, College of Mass Communication, University of the Philippines.

Pressman, R. S. (2010). *Software Engineering A Practitioner's Approach*. McGraw-Hill Higher Education.